HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** microservices, implementing RESTful APIs for data exchange between internal systems, handling approximately 5,000 requests per minute.
- Engineered **Python** scripts for data ingestion and transformation, creating automated ETL pipelines that processed over 10 GB of daily financial transaction data.
- Designed and optimized **SQL** queries across **PostgreSQL** and **Oracle** databases, improving data retrieval performance for critical reporting functions by **18%**.
- Contributed to the design and implementation of a quantitative trading strategy simulation, focusing on backtesting performance metrics and risk parameter tuning.
- Participated in weekly poker and strategy board game sessions to develop tactical decision-making skills and anticipate opponent moves in simulated trading scenarios.
- Analyzed and debugged complex issues within a high-frequency trading environment, identifying root causes and implementing corrective actions to minimize latency.
- Containerized key trading application components using **Docker** and deployed them to a managed Kubernetes cluster, ensuring consistent execution environments.

SKILLS

Languages – Python, Java, SQL, R

Frameworks/Libraries – Spring Boot, pandas, NumPy, scikit-learn, Flask, FastAPI

Databases – PostgreSQL, MySQL, Oracle

Cloud/DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux

Analytics/BI – Power BI, Tableau, DAX, Power Query, Data Modelling, Data Warehousing

Concepts – Game Theory, Options Trading, ETF Trading, Quantitative Research, Risk Management, Machine Learning Fundamentals

PROJECTS

Trading Strategy Backtesting Engine (HCL Technologies)

- Developed a **Python**-based backtesting framework to simulate trading strategies on historical market data, processing datasets exceeding 50GB.
- Implemented core logic for strategy evaluation, including calculation of key performance indicators such as Sharpe Ratio and Max Drawdown.
- Integrated a **FastAPI** service for real-time strategy parameter adjustment, enabling rapid iteration during the development cycle.

Market Data Analysis Dashboard (MSc Applied Project)

- Built an interactive **Power BI** dashboard to visualize and analyze real-time market data feeds, presenting trends and volatility metrics.
- Applied **DAX** measures and Power Query for data transformation and complex calculations, supporting analysis of over 100 distinct financial instruments.
- Collaborated with academic advisors to refine data presentation and ensure actionable insights for simulated trading decisions.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)